SHRI VAISHNAV VIDYAPEETH VISHWAVIDYALAYA, INDORE

Shri Vaishnav Institute of Information Technology (SVIIT) • Dept. of CSE

B.Tech. CSE Semester-VII (2026–2027) | BTCS706N Major Project (Practical 8 hrs/wk • 200 Marks)

WEEKLY PROGRESS REPORT — SPRINT 1 OF 6 (PHASE 1: INCEPTION & FUNCTIONAL MODELING)

Dates: August 10, 2026 – August 23, 2026 | **Team ID:** SVVV-CSE-2026-MP-14 | **Guide:** Prof. Shweta Pandey | **Status: Completed / On Track**

Project Title:	Development of e-Portal for Facilitating Case Management Hearing of Various Types of Cases (Order XV-A CPC)
Team Members:	1. Mukul Bhamgare (Lead Architect) 2. Rashi Kashiv (Frontend Lead) 3. Sakshi Singh (Data Architect) 4. Kanishk (DevOps / QA)
Sprint Goal:	Establish statutory system boundaries under Order XV-A CPC, configure Jira & Git repository with branch protections, and construct IEEE/Cockburn Use Case specifications (UC-01 to UC-06).

1. UML & OOAD Engineering Progress

• **Diagrams Completed:** Use Case Diagram (System Boundary enclosing UC-01 to UC-06 with 4 human and 3 system actors), Subsystem Functional Boundary Diagrams with <> (RBAC, SHA-256 Hashing, Conflict Detection) and <> (ADR Mediation, Deadline Alerts).

• **Diagrams Revised:** Actor Hierarchy Refinement (Segregated Advocate with e-filing/scheduling rights from Litigant with read-only tracking and alert rights; decoupled legacy CIS 3.2 / NJDG as external API boundary actors).

• **OO Concepts & Design Patterns:** Actor Generalization & Specialization (Base User actor specialized into JudicialOfficer, Advocate, CourtRegistryClerk, and Litigant with distinct RBAC boundaries); Functional Decomposition using <> and <>.

2. Technical Execution (DevOps, Git & Jira)

• **Jira Sprint Backlog:** [CMH-101] Git & Jira Scrum Board Initialization; [CMH-102] Statutory Workflow Specification under Order XV-A CPC; [CMH-103] Actor Hierarchy & Boundary Identification; [CMH-104] UML Use Case Diagram Construction; [CMH-105] IEEE Cockburn Formal Use Case Descriptions; [CMH-106] Requirements Traceability Matrix (RTM).

• **Git Feature Branches:** `feature/cmh-101-repo-scaffold`, `feature/cmh-104-use-case-boundary`, `feature/cmh-105-cockburn-specs`

• **Peer-Reviewed Pull Requests:** PR #01 (Repository scaffold & Jira workflow), PR #04 (Order XV-A use case models), PR #07 (Traceability matrix & Cockburn specs)

• **CI/CD Pipeline Status:** Build Passing | 100% Lint & Syntax Check (Flake8 & Prettier)

3. Automated Testing & Defect Tracking

• **Total Tests / Pass Rate: 12 Smoke & Validation Tests** (Pass Rate: **100%**)

• **Key Bug Identified & Overcome:** Discrepancy in statutory timeline limits between standard CPC Order X and Commercial Courts Act Order XV-A Rule 1 (resolved by parameterizing procedural rules into configurable statutory policy constants).

4. Goals for Next Week / Sprint

• [Goal A] Phase 2: Static & Dynamic Object-Oriented Analysis (OOA) — Model domain entities and multiplicities (1:1, 1:N, M:N); [Goal B] Construct Object Diagram for active commercial suit snapshot and Activity Diagram for Order XV-A trial lifecycle.

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Project Guide Signature
(Prof. Shweta Pandey)

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Project In-charge Signature
(Department of CSE, SVIIT)

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WEEKLY PROGRESS REPORT — SPRINT 2 OF 6 (PHASE 2: STATIC & DYNAMIC OBJECT-ORIENTED ANALYSIS)

Dates: August 24, 2026 – September 06, 2026 | **Team ID:** SVVV-CSE-2026-MP-14 | **Guide:** Prof. Shweta Pandey | **Status: Completed / On Track**

Project Title:	Development of e-Portal for Facilitating Case Management Hearing of Various Types of Cases (Order XV-A CPC)
Team Members:	1. Mukul Bhamgare (Lead Architect) 2. Rashi Kashiv (Frontend Lead) 3. Sakshi Singh (Data Architect) 4. Kanishk (DevOps / QA)
Sprint Goal:	Model domain entities, relations, runtime instance snapshots, and Activity workflows for pre-trial fact crystallization and admissions.

1. UML & OOAD Engineering Progress

• **Diagrams Completed:** Object Diagram (Runtime snapshot of Commercial Suit COMM_SUIT_301_2026 in Indore Bench), Activity Diagram (Pre-trial fact crystallization lifecycle with 120-day statutory default forfeiture partition).

• **Diagrams Revised:** Domain Model Multiplicities (Updated Case to AdmissionsMatrix relationship from 1:1 to 1:N history snapshotting to preserve audit trails of amended averments).

• **OO Concepts & Design Patterns:** State Modeling & Encapsulation (Modeled CaseLifecycleState transitions as encapsulated finite state machine with strict state guard assertions; applied Composition between Case and DocketPleadings).

2. Technical Execution (DevOps, Git & Jira)

• **Jira Sprint Backlog:** [CMH-201] Role-Based Access Control (RBAC) & JWT Security; [CMH-202] Docket Ingestion & SHA-256 Integrity Service; [CMH-203] Interactive Admissions & Denials Domain Model; [CMH-204] Judge Pre-Trial Summary View; [CMH-205] Advocate Multi-Bench Availability Domain Modeling; [CMH-206] Calendar Synchronization Service Prototype.

• **Git Feature Branches:** `feature/cmh-201-jwt-rbac`, `feature/cmh-202-sha256-vault`, `feature/cmh-203-admissions-domain`, `feature/cmh-205-scheduling-models`

• **Peer-Reviewed Pull Requests:** PR #11 (JWT authentication & password hashing), PR #14 (SQLAlchemy 2.0 ORM schemas & SQLite WAL), PR #16 (Admissions matrix domain model & state transitions)

• **CI/CD Pipeline Status:** Build Passing | 89.4% Code Coverage (GitHub Actions Pytest & Alembic migrations)

3. Automated Testing & Defect Tracking

• **Total Tests / Pass Rate: 28 Unit & Integration Tests** (Pass Rate: **100%**)

• **Key Bug Identified & Overcome:** Database connection pool exhaustion (QueuePool limit overflow) during concurrent advocate login simulations; resolved by reconfiguring SQLAlchemy session pool parameters (pool_size=20, max_overflow=10, pool_recycle=1800).

4. Goals for Next Week / Sprint

• [Goal A] Phase 3: Object-Oriented Design (OOD) — Construct Detailed Class Diagram with private attributes and typed method signatures; [Goal B] Formulate Sequence Diagram for admissions adjudication and Package Diagram for decoupled micro-architecture.

Project Guide Signature **(Prof. Shweta Pandey)**	**Project In-charge Signature** **(Department of CSE, SVIIT)**

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WEEKLY PROGRESS REPORT — SPRINT 3 OF 6 (PHASE 3: OBJECT-ORIENTED DESIGN & INTERACTION MODELING)

Dates: September 07, 2026 – September 20, 2026 | **Team ID:** SVVV-CSE-2026-MP-14 | **Guide:** Prof. Shweta Pandey | **Status:** Completed / On Track

Project Title:	Development of e-Portal for Facilitating Case Management Hearing of Various Types of Cases (Order XV-A CPC)
Team Members:	1. Mukul Bhamgare (Lead Architect) 2. Rashi Kashiv (Frontend Lead) 3. Sakshi Singh (Data Architect) 4. Kanishk (DevOps / QA)
Sprint Goal:	Detail Class Diagram, formulate dynamic message interactions via Sequence Diagram, structure modular Package Diagram, and build the Admissions & Denials Matrix engine.

1. UML & OOAD Engineering Progress

• **Diagrams Completed:** Detailed Class Diagram (Complete class hierarchy, attribute visibilities +/-, typed signatures, inheritance, and GoF patterns), Sequence Diagram (Pre-Trial CMH Workflow with MP Cyber Treasury and NJDG), Package Diagram (Decoupled package structure).

• **Diagrams Revised:** Decoupled HearingSchedule from direct Case entity dependency; introduced independent AdvocateAvailabilityManager to prevent tight table coupling in database.

• **OO Concepts & Design Patterns:** GoF Creational Design Patterns: Singleton Pattern (Double-checked locking JudicialVaultSingleton for cryptographic audit integrity and DatabaseConnectionPoolSingleton); Factory Method Pattern (CMODecreeFactory with Commercial, Summary, and Cheque Bounce concrete factories).

2. Technical Execution (DevOps, Git & Jira)

• **Jira Sprint Backlog:** [CMH-301] Class Diagram & ORM Entity Mapping; [CMH-302] Sequence Diagram & Message Interaction Formulation; [CMH-303] Package Diagram Modularization; [CMH-304] Admissions & Denials Engine API; [CMH-305] Readiness Score & Dispute Locking Mechanism; [CMH-306] Design Patterns Architecture Formulation.

• **Git Feature Branches:** `feature/cmh-301-class-diagram-entities`, `feature/cmh-304-admissions-matrix-api`, `feature/cmh-306-gof-patterns-singleton-factory`

• **Peer-Reviewed Pull Requests:** PR #21 (Admissions REST router & readiness engine), PR #25 (GoF Singleton & Factory implementations), PR #28 (Detailed UML sequence and class definitions)

• **CI/CD Pipeline Status:** Build Passing | 91.2% Code Coverage (Pytest with Swagger OpenAPI schema generation)

3. Automated Testing & Defect Tracking

• **Total Tests / Pass Rate: 42 Unit & Design Pattern Tests** (Pass Rate: **100%**)

• **Key Bug Identified & Overcome:** Concurrency race condition during simultaneous line-item admission submissions by opposing counsels; resolved via atomic SQLite transaction locks and SQLAlchemy row-versioning checks.

4. Goals for Next Week / Sprint

• [Goal A] Phase 3/4: System Architecture — Construct Component Diagram and Physical Deployment Diagram; [Goal B] Implement Strategy Design Pattern for advocate and courtroom collision detection.

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(Prof. Shweta Pandey)

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Project In-charge Signature
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WEEKLY PROGRESS REPORT — SPRINT 4 OF 6 (PHASE 3/4: SYSTEM ARCHITECTURE & CONFLICT SCHEDULING)

Dates: September 21, 2026 – October 04, 2026 | **Team ID:** SVVV-CSE-2026-MP-14 | **Guide:** Prof. Shweta Pandey | **Status: Completed / On Track**

Project Title:	Development of e-Portal for Facilitating Case Management Hearing of Various Types of Cases (Order XV-A CPC)
Team Members:	1. Mukul Bhamgare (Lead Architect) 2. Rashi Kashiv (Frontend Lead) 3. Sakshi Singh (Data Architect) 4. Kanishk (DevOps / QA)
Sprint Goal:	Construct Component & Deployment Diagrams, implement Clash-Free Hearing Scheduler using Strategy Pattern, and build Nginx reverse proxy Docker architecture.

1. UML & OOAD Engineering Progress

• **Diagrams Completed:** Component Diagram (5-tier decoupled micro-architecture: Presentation, Gateway, Business, Storage, External), Deployment Diagram (Physical node topologies, Uvicorn ASGI, SQLite WAL/MySQL, Nginx container, external Cyber Treasury gateways).

• **Diagrams Revised:** Deployment topology revised to support both 1-command Docker Compose orchestration (local evaluation) and Cloud PaaS topology (Render ASGI + Vercel SPA).

• **OO Concepts & Design Patterns:** GoF Behavioral Design Pattern: Strategy Pattern (Decoupled scheduling algorithms via SchedulingStrategy: StrictBenchSlottingStrategy, FlexibleBufferSlottingStrategy, and VirtualHybridSlottingStrategy hot-swapped via ClashDetectionContext).

2. Technical Execution (DevOps, Git & Jira)

• **Jira Sprint Backlog:** [CMH-401] Component Diagram & Service Decoupling; [CMH-402] Deployment Diagram & Cloud Infrastructure; [CMH-403] Strategy Pattern Clash Detection Engine; [CMH-404] Multi-Bench Calendar & Daily Cause List Router; [CMH-405] Docker Compose & Nginx Reverse Proxy Containerization; [CMH-406] GoF Swagger Catalog API.

• **Git Feature Branches:** `feature/cmh-403-scheduling-strategy`, `feature/cmh-404-cause-list-api`, `feature/cmh-405-docker-orchestration`

• **Peer-Reviewed Pull Requests:** PR #32 (Conflict-free scheduling router with strategy pattern), PR #36 (Docker, Docker Compose & Nginx configuration), PR #39 (Design pattern catalog & Swagger documentation)

• **CI/CD Pipeline Status:** Build Passing | 93.8% Code Coverage (Automated container health check & Pytest)

3. Automated Testing & Defect Tracking

• **Total Tests / Pass Rate: 56 Unit & Strategy Integration Tests** (Pass Rate: **100%**)

• **Key Bug Identified & Overcome:** False-positive courtroom clash detected during virtual video-conference (VC) hearings; resolved by introducing VirtualHybridSlottingStrategy where physical hall collisions are bypassed while advocate personal double-booking remains strictly prohibited.

4. Goals for Next Week / Sprint

• [Goal A] Phase 4: Implementation & Full-Stack Integration — Build interactive React 19 single-page application with role switcher; [Goal B] Implement Case Management Order (CMO) Sheet with QR code verification and SHA-256 seal.

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WEEKLY PROGRESS REPORT — SPRINT 5 OF 6 (PHASE 4: FULL-STACK INTEGRATION, CMO ENGINE & E-CHALLAN)

Dates: October 05, 2026 – October 18, 2026 | **Team ID:** SVVV-CSE-2026-MP-14 | **Guide:** Prof. Shweta Pandey | **Status: Completed / On Track**

Project Title:	Development of e-Portal for Facilitating Case Management Hearing of Various Types of Cases (Order XV-A CPC)
Team Members:	1. Mukul Bhamgare (Lead Architect) 2. Rashi Kashiv (Frontend Lead) 3. Sakshi Singh (Data Architect) 4. Kanishk (DevOps / QA)
Sprint Goal:	Integrate React 19 judicial dashboard with FastAPI backend, implement Case Management Order (CMO) sheet with SHA-256 cryptographic seal, 16-digit CNR lookup, and MP Cyber Treasury court fees.

1. UML & OOAD Engineering Progress

• **Diagrams Completed:** Runtime Object Collaboration Model (Sealed CMO decree formulation and digital QR verification pipeline), Updated Sequence Diagram incorporating e-Challan payment gateway callbacks.

• **Diagrams Revised:** Frontend layout restructured from standard tabs to multi-role judicial persona views (Judge, Advocate, Clerk, Citizen/Litigant) to satisfy SVVV RBAC evaluation criteria.

• **OO Concepts & Design Patterns:** Polymorphic Decree Generation (CMODecreeFactory generates specialized decree models according to case category); Inversion of Control & Dependency Injection (FastAPI Depends for isolated database sessions and auth contexts).

2. Technical Execution (DevOps, Git & Jira)

• **Jira Sprint Backlog:** [CMH-501] Full-Stack CI/CD Pipeline Automation; [CMH-502] Case Management Order (CMO) Sheet & QR Seal Engine; [CMH-503] React 19 Judicial Dashboard & Role Switcher; [CMH-504] 16-Digit CNR Search & Filter Engine; [CMH-505] MP Cyber Treasury e-Challan & Fee Verification Module; [CMH-506] Multi-Channel Notification Router (SMS/WhatsApp/In-App).

• **Git Feature Branches:** `feature/cmh-502-cmo-qr-engine`, `feature/cmh-503-react-judicial-ui`, `feature/cmh-505-echallan-cyber-treasury`

• **Peer-Reviewed Pull Requests:** PR #44 (React 19 judicial UI with dark/light themes), PR #48 (CMO generation with SHA-256 cryptographic seal & QR stub), PR #51 (e-Challan & multi-channel notification engine)

• **CI/CD Pipeline Status:** Build Passing | Vite v8.3.1 Clean Build (0 errors) & 100% Pytest Green

3. Automated Testing & Defect Tracking

• **Total Tests / Pass Rate: 27 Core Suite Tests (100% Pass)** (Pass Rate: **100%**)

• **Key Bug Identified & Overcome:** Cryptographic checksum mismatch between backend SHA-256 hash generation and client-side canvas QR code payload; resolved by standardizing canonical JSON serialization (sorted keys, compact delimiters) across Python and JavaScript.

4. Goals for Next Week / Sprint

• [Goal A] Phase 5: Testing, Cloud Deployment & Final Defense — Execute stress tests, deploy to public cloud (Render & Vercel), finalize A4 documentation dossiers, and rehearse oral viva voce defense.

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WEEKLY PROGRESS REPORT — SPRINT 6 OF 6 (PHASE 5: AUTOMATED TESTING, CLOUD DEPLOYMENT & VIVA DEFENSE)

Dates: October 19, 2026 – November 01, 2026 | **Team ID:** SVVV-CSE-2026-MP-14 | **Guide:** Prof. Shweta Pandey | **Status: Completed / Final Review Ready**

Project Title:	Development of e-Portal for Facilitating Case Management Hearing of Various Types of Cases (Order XV-A CPC)
Team Members:	1. Mukul Bhamgare (Lead Architect) 2. Rashi Kashiv (Frontend Lead) 3. Sakshi Singh (Data Architect) 4. Kanishk (DevOps / QA)
Sprint Goal:	Complete automated testing with 100% pass rate, deploy portal to public cloud URL (Vercel & Render), compile print-ready A4 project report and UML dossier, and prepare oral viva defense presentation.

1. UML & OOAD Engineering Progress

• **Diagrams Completed:** Print-Ready 8-Page A4 PDF Dossier (SVVV_UML_Architecture_Dossier_A4.pdf) containing all 7 standard UML diagrams with official university cover sheet; Interactive HTML Print Dossier (Printable_UML_Book.html).

• **Diagrams Revised:** All 7 UML diagrams verified against actual code implementation (100% UML-to-code consistency verified for Class, Sequence, and Deployment diagrams).

• **OO Concepts & Design Patterns:** Complete OOAD Realization: Encapsulation (private members and getters), Inheritance & Polymorphism (Strategy and Factory hierarchies), SOLID principles, and 3 GoF Design Patterns verified with unit tests.

2. Technical Execution (DevOps, Git & Jira)

• **Jira Sprint Backlog:** [CMH-601] Automated Unit & Integration Test Suites (Pytest 27 passed); [CMH-602] Negative Path & Concurrency Stress Testing; [CMH-603] Cloud Staging & Production Deployment (Render + Vercel); [CMH-604] System Usability Scale (SUS > 85) & Latency Benchmark; [CMH-605] Academic Final Project Report (BTCS706N 50KB+); [CMH-606] Viva Voce Defense Deck (16 Slides HTML) & Script.

• **Git Feature Branches:** `develop, feature/cmh-601-test-isolation, feature/cmh-603-cloud-deployment, main`

• **Peer-Reviewed Pull Requests:** PR #55 (GoF design pattern tests & coverage report), PR #58 (A4 PDF dossier generator & printable report), PR #60 (Cloud deployment configuration for Render and Vercel)

• **CI/CD Pipeline Status:** Build Passing | All 27 Pytest Tests Green | Vite Production Build Clean (191ms)

3. Automated Testing & Defect Tracking

• **Total Tests / Pass Rate: 27 Pytest Tests (100% Pass Rate)** (Pass Rate: **100%**)

• **Key Bug Identified & Overcome:** Test database mutation during pytest runs contaminating the local development database; resolved by introducing isolated temporary SQLite test engine fixtures with automatic setup and teardown in conftest.py.

4. Goals for Next Week / Sprint

• [Goal A] Conduct final external examiner evaluation and live portal demonstration; [Goal B] Archive project artifacts, repository tag v1.0.0, and hardcopy spiral-bound report submission to SVIIT CSE department.

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Project Guide Signature **(Prof. Shweta Pandey)**	**Project In-charge Signature** **(Department of CSE, SVIIT)**

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